D Y PATIL INTERNATIONAL UNIVERSITY
Sector 29, Pradhikaran, Akurdi, Pune - Maharashtra, INDIA 411044
(Establishment by Maharashtra Act No. LXIII of 2017)
School of Computer Science, Engineering & Applications

Course Name	Basic Electrical and Electronics Engineering
Course Code	CSE1002
Credits	3
Prerequisites	Basic knowledge of Physics, Elementary Mathematics
L-T-P-C	3-0-0-3

COURSE OBJECTIVES

1. To understand the fundamental concepts of electrical quantities, circuit elements, sources, and basic electrical circuits.
2. To explain the working principles and applications of semiconductor devices, diodes, rectifiers, and Zener diode voltage regulators.
3. To apply basic electrical and electronic concepts to interface sensors and Arduino
4. To analyze the operation of BJT and MOSFET as electronic switches
5. To implement basic operational amplifier circuits and understand their applications in amplification

COURSE SYLLABUS

Modules	**No. of Lectures**
Module I: Fundamentals of Electrical Circuits Introduction to electrical engineering, Basic electrical quantities: voltage, current, power and energy, Circuit elements: resistor, inductor and capacitor, independent voltage and current sources, Series and parallel circuits, Ohm's Law, Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL)	6
Module II: Semiconductor Devices and Applications Introduction to semiconductors: conductors, insulators and semiconductors, P-type and N-type semiconductors, PN junction diode: construction, working principle and V-I characteristics, Diode as a switch, Rectifiers: Half-wave and Full-wave/Bridge – working and applications, Zener diode as a voltage regulator.	8
Module III- Sensors and Arduino Applications Introduction to sensors and transducers, classification of sensors: analog and digital sensors, Applications of sensors in everyday life and computer engineering, Introduction to Arduino, Sensor Interfacing with Arduino: Interfacing LED and push button with Arduino, Interfacing LDR/light sensor, interfacing temperature sensor, Interfacing soil moisture sensor	8

Module IV: Bipolar Junction Transistors (BJT) and MOSFET Introduction to BJT: construction, terminals and working principle, BJT characteristics, BJT as an electronic switch, BJT as an amplifier, Common Emitter, Common Base and Common Collector configurations, Introduction to MOSFET, N-channel and P-channel MOSFET – basic operation, MOSFET as an electronic switch, Comparison of BJT and MOSFET.	8
Module V: Operational Amplifiers and Applications Introduction to operational amplifier, differential amplifier and ideal Op-Amp characteristics, inverting and non-inverting amplifier configurations, voltage follower, adder and subtractor, Op-Amp as comparator, OP-AMP as integrator and differentiator.	6
Self Study / Concurrent Assessment Hours	**9**
Total Contact Hours (36+9)	**45**

COURSE OUTCOMES At the end of the course the student will be able to:	
CO1	Explain the fundamental concepts of electrical quantities, circuit elements, and sources, and apply Ohm's Law, KCL, and KVL to analyze and solve basic electrical circuits.
CO2	Explain the basic characteristics and working principles of semiconductor devices and apply PN junction and Zener diodes in switching, rectification, and voltage regulation circuits.
CO3	Explain sensors and Arduino and apply basic sensor interfacing techniques for real-world applications.
CO4	Analyze BJT and MOSFET characteristics and apply them as electronic switches and amplifiers
CO5	Explain the basic characteristics of operational amplifiers and apply Op-Amp configurations for signal amplification and basic mathematical operations.

CO-PO Mapping:	3: Substantial			2: Moderate				1: Slight					
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	3	2	–	–	–	1	–	–	1	2	2
CO2	3	3	2	3	2	–	–	–	1	–	–	1	2	2
CO3	3	2	3	2	3	1	1	–	2	1	–	2	2	3
CO4	3	3	2	3	2	–	–	–	1	–	–	1	2	3
CO5	3	3	3	2	2	–	–	–	1	1	–	2	3	3

TEXTBOOKS

1. Thomas L. Floyd, "Electronic Devices",9th-edition, Pearson.
2. V.K. MEHTA, ROHIT MEHTA Publisher, "Principles of Electrical Engineering and Electronics" S Chand.
3. Charles K. Alexander _9850,Fundamentals of Electric Circuits (Alexander and Sadiku), 4th Edition.

REFERENCES

1. David A. Bell, “Electronic Devices and Circuits”, Oxford University Press, 5th Edition, 2008.
2. D.P. Kothari, I. J. Nagrath, “Basic Electronics”, McGraw Hill Education (India) Private Limited, 2014.
3. Arduino Cookbook – Michael Margolis, Brian Jepson & Nicholas Robert O'Brien, O'Reilly

EVALUATION SCHEME

Theory: 60%			
Continuous Assessment	**Mid-term**	**End-term**	**Total**
25	25	50	100

Course Name	CSE1002P
Course Code	Basic Electrical and Electronics Engineering
Credits	1
Prerequisites	Basic knowledge of Physics, Elementary Mathematics
L-T-P-C	0-0-2-1

COURSE OBJECTIVES

1. To identify and measure basic electrical and electronic components and understand their practical characteristics.
2. To analyze and verify basic electrical circuits using Ohm's Law, KVL, and KCL, and understand diode, rectifier, and transistor applications.
3. To develop practical skills in Arduino UNO programming and sensor interfacing using digital and analog inputs.
4. To implement basic Op-Amp amplifier and signal-processing circuits and verify their operation experimentally.
5. To integrate electrical, electronic, Arduino, and sensor concepts to design and implement a small application-oriented mini project.

LABORATORY

Sr. No.	Content
1	Identification and Measurement of Basic Electronic Components
2	Verification of Kirchhoff's Laws (KVL & KCL) – Apply KVL in a series circuit and KCL in a parallel circuit, and verify through measurements.
3	Diode as a Switch and Zener Diode as Voltage Regulator
4	Study of Half-Wave Rectifier – Observe input and output waveforms, and calculate ripple factor.
5	Study of Full-Wave Rectifier (Center-tap & Bridge) – Compare performance of full-wave rectifiers with half-wave rectifier.
6	Introduction to Arduino UNO – Familiarization with Arduino UNO, Arduino IDE, digital/analog pins and basic programming; interface an LED and push button.
7	LDR-Based Light Sensing using Arduino – Interface an LDR with Arduino and control an LED based on light intensity.
8	Basic Op-Amp Applications – Implement and verify inverting, non-inverting amplifier and voltage follower circuits.
9	BJT as a Switch – Demonstrate switching action of a transistor using LED or small load.
10	Mini project

COURSE OUTCOMES

At the end of the course the student will be able to:

CO1	Identify and measure basic electrical and electronic components and verify fundamental circuit laws.
CO2	Analyze and implement diode, rectifier, Zener diode, and BJT-based circuits for practical
CO3	Program Arduino UNO and interface LEDs, push buttons, and sensors for basic automation applications.
CO4	Implement and verify basic Op-Amp circuits for amplification and signal processing.
CO5	Explain the basic characteristics of operational amplifiers and apply Op-Amp configurations for signal amplification and basic mathematical operations.

CO-PO Mapping:	3: Substantial			2: Moderate				1: Slight					
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	-	-	-	-	-	-	1	2	1
CO2	3	2	2	-	2	-	-	-	-	-	-	1	2	2
CO3	3	3	3	2	2	-	-	-	-	-	1	1	2	3
CO4	3	3	3	2	3	-	-	-	1	1	1	1	2	3
CO5	2	2	3	-	3	1	-	-	2	2	2	3	3	3

TEXTBOOKS

1. “Electronic-devices”-by-floyd-9th-edition.Prentice Hall
2. "Principles of Electrical Engineering and Electronics" by V.K. MEHTA, ROHIT MEHTA Publisher S Chand
3. Charles K. Alexander _9850Fundamentals of Electric Circuits (Alexander and Sadiku), 4th Edition.

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3. Arduino Cookbook – Michael Margolis, Brian Jepson & Nicholas Robert O'Brien, O'Reilly

Practical: 40%				
Continuous Assessment			**End-term Practical**	**Total**
Lab Manual	**Lab Assessment**	**Internal Viva-voce**		
10	10	30	50	100